Daycare Fees: Coverage Analysis

Lab 6 - Code Coverage and Branch Coverage

Project: L6_L7_48 - Daycare Fees

Objective: Measure test completeness, identify untested paths, and exercise the edge cases and decision branches in the daycare-fees application.

1. Code Coverage Analysis

Code coverage measures whether executable statements are run by the tests. The dedicated code-coverage suite tests invalid age groups and hours, included-hour boundaries, sibling allowances, age-specific fee rates, enrolment eligibility, and pickup statuses.

The coverage report contains 37 statements. All 37 statements were executed and no statements were missing, resulting in **100% line coverage** for `daycare_fees.py`.

Command used: `coverage run --source=. -m pytest CodeCoverageTest_daycare_fees.py`, followed by `coverage report -m`.

```
$ python3 -m coverage run --source=. -m pytest CodeCoverageTest_daycare_fees.py
.........                                                                [100%]
9 passed in 0.06s
$ python3 -m coverage report -m --include=daycare_fees.py
Name              Stmts   Miss  Cover   Missing
-----------------------------------------------
daycare_fees.py      37      0   100%
-----------------------------------------------
TOTAL                37      0   100%
```

Figure 1. Terminal output showing 100% code/statement coverage.

2. Branch Coverage Analysis

Branch coverage verifies both outcomes of each decision, rather than only checking whether a line was reached. The branch suite covers validation outcomes, sibling and age-group conditions, fee thresholds, enrolment conditions, aftercare and Friday cutoffs, and on-time, late, and overdue pickup outcomes.

The report contains 26 measured branches. All 26 branches were covered, with 0 partial branches, resulting in **100% branch coverage** and **100% statement coverage** for `daycare_fees.py`.

Command used: `coverage run --branch --source=. -m pytest BranchCoverageTest_daycare_fees.py`, followed by `coverage report -m`.

```
$ python3 -m coverage run --branch --source=. -m pytest BranchCoverageTest_daycare_fees.py
..................                                                       [100%]
18 passed in 0.06s
$ python3 -m coverage report -m --include=daycare_fees.py
Name              Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------
daycare_fees.py      37      0     26      0   100%
-------------------------------------------------------------
TOTAL                37      0     26      0   100%
```

Figure 2. Terminal output showing 100% branch coverage.

3. Test Deliverables and Conclusion

Deliverable	Result
Code coverage tests	CodeCoverageTest_daycare_fees.py
Branch coverage tests	BranchCoverageTest_daycare_fees.py
Code coverage result	37/37 statements, 100%
Branch coverage result	26/26 branches, 100%
Validation	29 tests passed in the complete Lab 6 suite

Together, the two suites demonstrate that all executable statements and all measured decision paths in the daycare-fees module are exercised by tests. This provides strong confidence that the tested edge cases and alternate outcomes are not left unverified.